

Aurora Innovation

NASDAQ : AUR · Young-Company DCF (Damodaran)

\$7.74
\$13.2B mkt cap · 1.71B sh · \$1.28B cash, \$650M net debt

THE CENTRAL QUESTION

Will Aurora capture meaningful share of a \$160B US autonomous trucking TAM by 2036 — or is today's \$13B market cap pricing certainty about driverless commercial economics that hasn't been proven?

Aurora trades at ~\$13.2B against \$11M of Q1 2026 revenue and \$640M annualized cash burn. The \$1.28B liquidity position needs replenishment by mid-2027 absent commercial conversion. Driverless commercial launch slipped from 2024 to now 2027; the Driver runs supervised highway pilots with PACCAR, Volvo, FedEx. Young-company DCF below: TAM share, terminal margin, P(failure). Three scenarios, weighted; show your work.

PROBABILITY-WEIGHTED EXPECTED VALUE

\$6.65
expected fair value today
-14.0% vs spot \$7.74

FORWARD COMPOUNDED VALUE

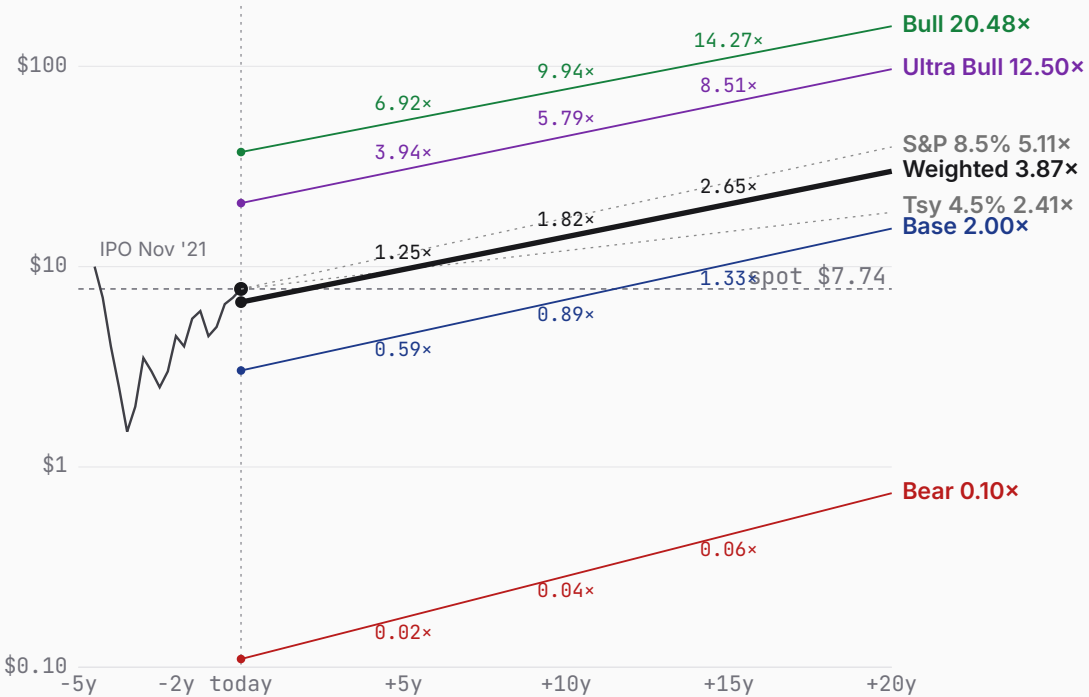
+5y	+10y	+15y	+20y
\$9.68	\$14.10	\$20.54	\$29.93
1.25x spot	1.82x spot	2.65x spot	3.87x spot

WEIGHTING RATIONALE

- Bear 35%** Autonomous trucking unproven at scale; Waymo Via shutdown signal still recent.
- Base 50%** Modal outcome: Aurora is one of 2-3 winners, 7-8% TAM share, tech-auto blend margins.
- Bull 12%** Compound conditional chain; software-take-rate platform thesis requires multiple alignments.
- Ultra Bull 3%** Tail of tails: dominant US autonomous + early software-take-rate + multiple expansion.

Bull (12%) + Ultra Bull (3%) together contribute \$5.10 — 77% of the \$6.65 expected value despite 15% combined probability. Today's spot (\$7.74) sits 16% above the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED FORWARD



BEAR	\$0.11	BASE	\$3.03	BULL	\$37.32	ULTRA BULL	\$20.76
	-98.6% vs spot		-60.9% vs spot		+382.2% vs spot		+168.2% vs spot
TAM 1.5% P(fail) 45% S2C 1.2 Prob 35%		TAM 7.5% P(fail) 25% S2C 1.5 Prob 50%		TAM 22.5% P(fail) 12% S2C 1.8 Prob 12%		TAM 6.6% P(fail) 3% S2C 2.5 Prob 3%	
Tech doesn't scale, capital exhausts.		Commercial scale works at limited share.		Dominant US platform, software-like margins.		Dominant US; software-take-rate.	



Aurora Innovation scenario narratives

PROBABILITY WEIGHTS

Bear 35% Base 50% Bull 12% Ultra Bull 3% · Weighted expected \$6.65 (-14.0% vs spot)

BEAR	\$0.11	BASE	\$3.03	BULL	\$37.32	ULTRA BULL	\$20.76
	-98.6% vs spot		-60.9% vs spot		+382.2% vs spot		+168.2% vs spot

Probability 35%

Tech doesn't scale, capital exhausts.

WHY 35%

Bearish thesis is compound: tech doesn't scale from supervised highway pilots to driverless at necessary safety thresholds; capital exhausts before commercial validation; dilution at falling prices accelerates as capital needs persist; competitive losses to Kodiak (cargo-focused, well-funded) and Waabi (smaller but capital-efficient). Within-scenario 45% p_fail captures Aurora-specific failure; the 35% scenario weight reflects category-level risk — autonomous trucking economics may not scale for Western operators. Waymo Via's 2023 shutdown was a credible signal of category difficulty. The 35% weight is slightly above JOBY's 30% bear because AUR is earlier-stage (pre-commercial), more capital-dependent, and faces more direct competition.

WHAT HAPPENS

The pessimistic case is brutal. Highway autonomy is harder than demos suggest — five years of 'next year' promises (launch was 2025, then 2026, now 2027), and the long tail of edge cases (night construction, weather, urban surface streets) proves more durable than current systems handle. Commercial routes stay limited to highway-only, daytime, sunbelt-only operation.

Cash burn doesn't moderate. Q1 2026's \$640M annualized outflow keeps rising. The \$1.28B liquidity needs replenishment by mid-2027. Successive raises come at lower prices — cumulative ~74% dilution by FY36, issuance averaging below \$3, destroying per-share value even when the business survives.

Probability 50%

Commercial scale works at limited share.

WHY 50%

Modal outcome. Aurora launches commercial Driverless service on schedule (2027), proves cost-per-mile advantage on a single route (Dallas-Houston initially), expands to 2-3 routes by 2029, secures pilot partnerships with major carriers (Werner, Schneider, FedEx). Reaches ~7-8% share of \$160B addressable autonomous trucking TAM by 2036 — meaningful but not dominant. Mature operating margin lands at ~20% (tech-auto blend, well below software economics). The 50% weight reflects that this is the modal outcome — most realistic for a strong technical team executing on a long technical-commercial timeline, but not requiring breakthrough commercial dominance.

WHAT HAPPENS

The Aurora Driver works at commercial scale. By FY28 the company operates 1,000–2,000 driverless trucks on commercial highways. McLane (Berkshire) expands beyond Texas. FedEx, Schneider, Werner pilots convert to multi-year volume commitments. Hirschbach's 500-truck order ships through 2027–28.

But the market splinters. Kodiak owns industrial off-highway. Waabi grows through the Volvo OEM channel. PlusAI lands legacy retrofits. Aurora becomes one of 2–3 'winners' in US for hire long-haul, capturing ~7.5% of the \$160B TAM by FY36 — \$12B revenue. Mature margin settles around 20% — a blend of software-licensing and own-fleet operations.

Probability 12%

Dominant US platform, software-like margins.

WHY 12%

Compound bet: Driverless commercial launches on schedule AND scaled trucking fleet ramps to 5,000+ trucks by 2028 AND software-take-rate model emerges (Aurora-as-platform on partner-operated fleet) AND competitive narrowing (Kodiak underperforms, Waabi stays niche) AND autonomous trucking proves uniquely viable. Each condition independently plausible at 40-60% probability; conjunctive probability lands at 15%. The asymmetric venture-bet pattern characteristic of autonomous-platform investments — bull case pays multiples-of-current- valuation if it converges, but requires conditions to align over 5-10 years.

WHAT HAPPENS

Aurora becomes the dominant US autonomous trucking platform. FY27 driverless commercial launch proceeds on schedule, unlocking follow-on orders from major carriers (Werner, Schneider, FedEx, Hirschbach). By FY28 fleet exceeds 5,000 trucks; by FY31, 50,000+. Network effects compound — lane data, edge cases, regulatory relationships in TX/OK/NM/AZ.

The competitive set narrows. Kodiak stays viable but smaller. Waabi struggles. PlusAI runs out of SPAC capital. By FY32 only Aurora and one other player operate at-scale driverless commercial fleets. Aurora captures 22.5% of the \$160B TAM — \$36B FY36 revenue. Pivot to software-take-rate on OEM trucks (Volvo VNL, PACCAR). Mature margin reaches 35% — similar to Mobileye's ADAS economics.

Probability 3%

Dominant US; software-take-rate.

WHY 3%

Tail of tails: commercial launch on schedule (50%) AND Aurora Driver becomes the default software on PACCAR/Volvo fleets reaching ~20% of autonomous miles / ~6–7% take-rate share of TAM by 2036 (30% conditional on launch) AND software-take-rate model wins (40%) AND multiple expansion to software/aerospace-hybrid premium (50%). Each individually plausible; joint at 3-4%. Captures the scenario where Aurora becomes a category-defining Big Tech-tier company.

WHAT HAPPENS

The ultra-bull case is that Aurora becomes the dominant US autonomous trucking platform AND wins the software-take-rate game. Commercial launch on schedule by 2027 + Aurora Driver reaches 1M+ autonomous miles by 2028 + Aurora Driver running on ~20% of US autonomous long-haul miles by 2036, monetized via per-mile take-rate (~6–7% revenue-share of TAM) with PACCAR/Volvo/FedEx fleet partners rather than capex-heavy direct ownership. By FY36 Aurora generates \$8B+ revenue at 40%+ software-like margins.

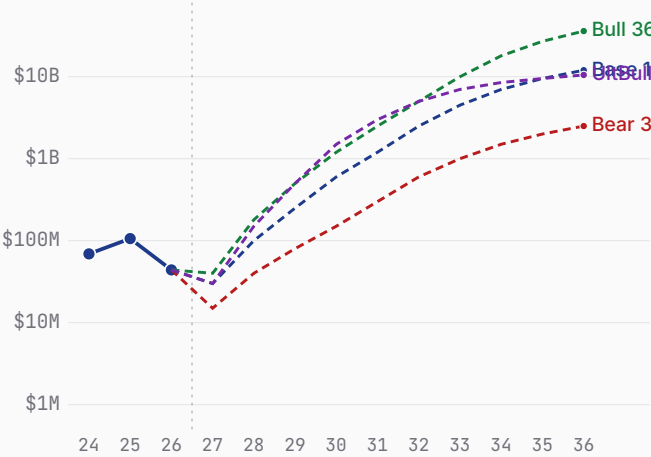
Compound conditional: each piece individually plausible at 50-65%; joint at 3-4%. Reflected here at 3%. This is the scenario where Aurora joins the Tesla/Nvidia tier — not just 'autonomy works,' but 'Aurora wins the autonomy market.' Contributes \$3.00 to weighted (\$100 × 3%) — the asymmetric tail.

Aurora Innovation business snapshot

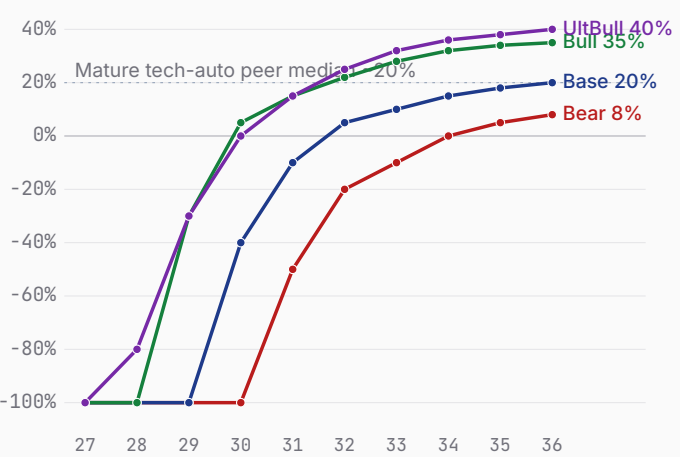
Bear \$0.11 Base \$3.03 Bull \$37.32

FY24–FY26 history + FY27–FY36 scenario projections · fiscal years end Dec 31 · 10-K, Q1 2026 10-Q, ATA/FTR Tonnage Index

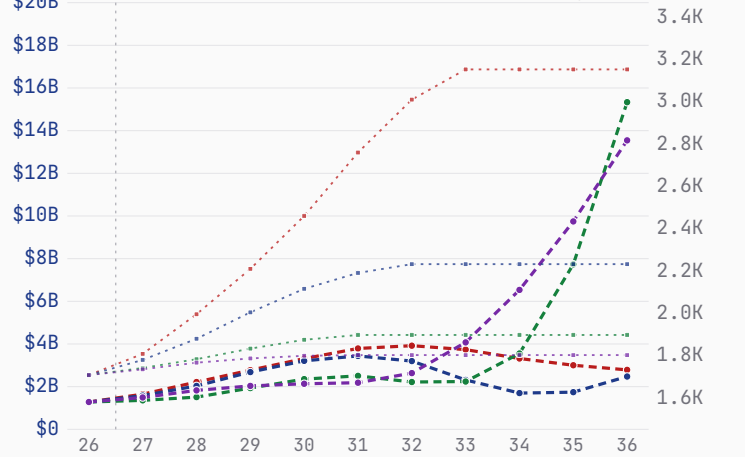
Revenue — history + scenarios to FY36 (log)



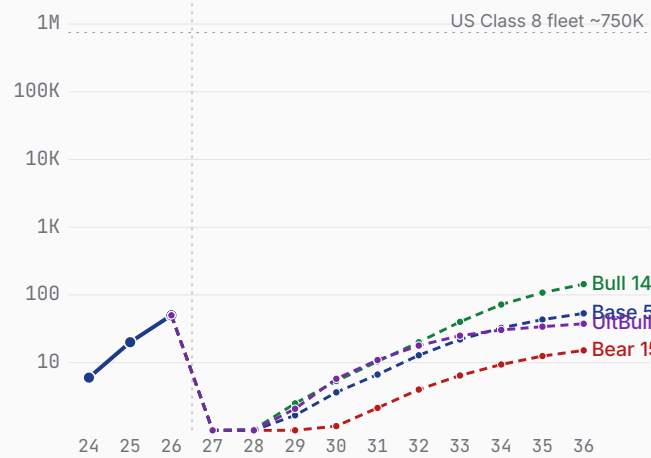
Path to profitability — op margin (FY27→FY36)



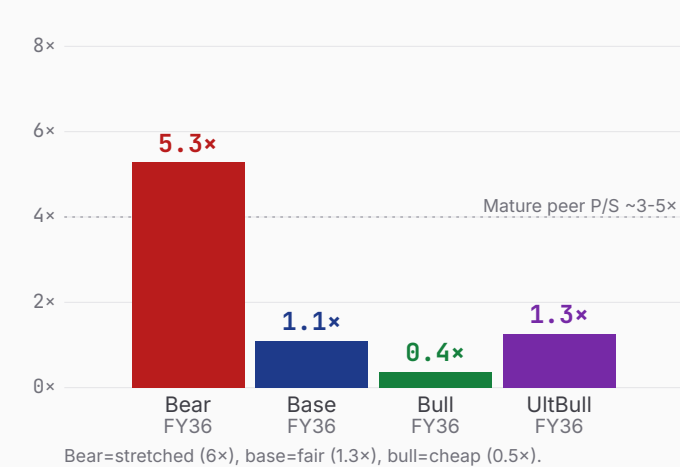
Cash + dilution (FY26 → FY36)



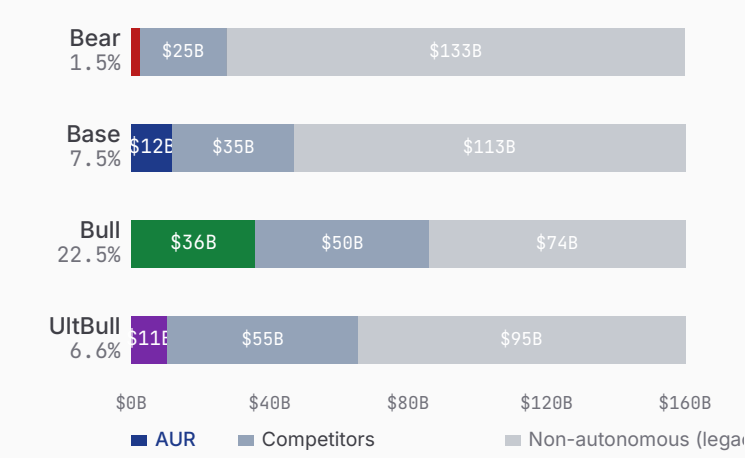
Driverless trucks deployed (log)



\$13.20B mkt cap as P/S on FY36 rev



\$160B US autonomous trucking TAM (FY36) — AUR share



Sources: AUR 10-K FY25, Q1 2026 10-Q, ATA/FTR Tonnage Index (US for-hire long-haul \$300B; autonomy-addressable \$160B).

Aurora Innovation 7 Powers · the moat, scored

Bear \$0.11 Base \$3.03 Bull \$37.32

Arena. Driverless Class-8 trucking on US highways — Aurora vs Kodiak, Waabi, Torc/Daimler, Gatik; the race to scale commercial driverless freight.

POWER ORIENTATION
origination window: open

POWER ORIENTATION · 7 POWERS

Scale Economies	Per-mile cost falls with fleet, but the driverless fleet is barely launched; R&D is still a fixed sink.
Network Economies	Aurora Driver improves with miles driven; a data flywheel, not yet a moat.
Counter-Positioning	Driver-as-a-service subscription model carriers/OEMs can't easily self-build.
Switching Costs	Carrier/terminal integration nascent; lane-by-lane.
Branding	Safety-case credibility; not yet the category default.
Cornered Resource · thesis leans here	Aurora Driver + FirstLight lidar IP; PACCAR/Volvo/Continental platform partnerships; Dallas-Houston-Phoenix terminals.
Process Power	Safety Case Framework rigor; not yet a durable cost/quality edge.

THE RACE · NAMED RIVALS

Kodiak	SPAC; ~\$ hundreds M
~12% terminal share · Driverless in the Permian (off-highway first); SPAC-funded.	
Waabi	~\$200M raised
~10% terminal share · Generative-AI simulation approach; Volvo/Khosla-backed.	
Torc Robotics (Daimler)	Daimler balance sheet
~10% terminal share · OEM-captive; Freightliner integration.	
Gatik	~\$300M raised
~8% terminal share · Short-haul B2B (Walmart, Tyson); narrower problem.	

LEAD / LAG · FALSIFIERS

Driverless commercial launch	live (TX) vs off-highway	LEADING
OEM platform partners	3 vs 1	LEADING
Cash runway (yrs at burn)	2 vs 3	LAGGING

COMPETITIVE READ

A genuine first-to-driverless and OEM-partnership lead, but scale and network power are unbuilt and the model is deeply capital-hungry; the window is open while Aurora out-executes — a safety or capital stumble cedes it.

Aurora Innovation show your work

Bear \$0.11 Base \$3.03 Bull \$37.32 · Weighted \$6.65 (-14.0%)

BEAR					BASE					BULL					ULTRA BULL				
\$0.11					\$3.03					\$37.32					\$20.76				
ASSUMPTIONS					ASSUMPTIONS					ASSUMPTIONS					ASSUMPTIONS				
TAM Year-10 (\$B)		160			TAM Year-10 (\$B)		160			TAM Year-10 (\$B)		160			TAM Year-10 (\$B)		160		
Mature share (%)		1.5			Mature share (%)		7.5			Mature share (%)		22.5			Mature share (%)		6.6		
Mature op margin (%)		8			Mature op margin (%)		20			Mature op margin (%)		35			Mature op margin (%)		40		
Sales-to-capital		1.2			Sales-to-capital		1.5			Sales-to-capital		1.8			Sales-to-capital		2.5		
Terminal WACC (%)		10.0			Terminal WACC (%)		8.5			Terminal WACC (%)		7.5			Terminal WACC (%)		8.0		
Terminal growth (%)		2.0			Terminal growth (%)		2.5			Terminal growth (%)		3.0			Terminal growth (%)		3.5		
P(failure) (%)		45			P(failure) (%)		25			P(failure) (%)		12			P(failure) (%)		3		
Scenario probability (%)		35			Scenario probability (%)		50			Scenario probability (%)		12			Scenario probability (%)		3		
10-YEAR DCF · \$B					10-YEAR DCF · \$B					10-YEAR DCF · \$B					10-YEAR DCF · \$B				
FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF
FY27	0.01	-800%	-0.13	-0.11	FY27	0.03	-600%	-0.20	-0.18	FY27	0.04	-500%	-0.22	-0.20	FY27	0.03	-250%	-0.10	-0.09
FY28	0.04	-400%	-0.18	-0.14	FY28	0.10	-250%	-0.30	-0.24	FY28	0.18	-150%	-0.35	-0.28	FY28	0.15	-80%	-0.24	-0.19
FY29	0.08	-200%	-0.19	-0.13	FY29	0.25	-100%	-0.35	-0.25	FY29	0.50	-30%	-0.33	-0.24	FY29	0.50	-30%	-0.50	-0.36
FY30	0.15	-100%	-0.21	-0.12	FY30	0.60	-40%	-0.47	-0.30	FY30	1.20	+5%	-0.33	-0.22	FY30	1.50	+0%	-0.75	-0.49
FY31	0.30	-50%	-0.28	-0.15	FY31	1.20	-10%	-0.52	-0.30	FY31	2.50	+15%	-0.35	-0.21	FY31	3.00	+15%	-0.75	-0.44
FY32	0.60	-20%	-0.37	-0.17	FY32	2.50	+5%	-0.77	-0.41	FY32	5.00	+22%	-0.29	-0.16	FY32	5.00	+25%	-0.25	-0.14
FY33	1.00	-10%	-0.43	-0.18	FY33	4.50	+10%	-0.98	-0.47	FY33	10.00	+28%	+0.02	+0.01	FY33	7.00	+32%	+0.84	+0.41
FY34	1.50	+0%	-0.42	-0.16	FY34	7.00	+15%	-0.84	-0.37	FY34	18.00	+32%	+1.32	+0.60	FY34	8.50	+36%	+1.81	+0.81
FY35	2.00	+5%	-0.32	-0.11	FY35	9.50	+18%	-0.32	-0.13	FY35	27.00	+34%	+4.18	+1.76	FY35	9.50	+38%	+2.61	+1.08
FY36	2.50	+8%	-0.22	-0.07	FY36	12.00	+20%	+0.23	+0.09	FY36	36.00	+35%	+7.60	+2.97	FY36	10.50	+40%	+3.45	+1.33
Σ PV explicit FCF				-1.35	Σ PV explicit FCF				-2.57	Σ PV explicit FCF				+4.02	Σ PV explicit FCF				+1.93
+ PV terminal (g 2.0%)				0.72	+ PV terminal (g 2.5%)				10.89	+ PV terminal (g 3.0%)				85.38	+ PV terminal (g 3.5%)				40.40
EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE				
Operating EV				\$-0.63B	Operating EV				\$8.32B	Operating EV				\$89.40B	Operating EV				\$42.33B
+ Cash & investments				\$1.28B	+ Cash & investments				\$1.28B	+ Cash & investments				\$1.28B	+ Cash & investments				\$1.28B
= Equity value				\$0.65B	= Equity value				\$9.60B	= Equity value				\$90.68B	= Equity value				\$42.96B
Raised over period				\$4.25B	Raised over period				\$4.50B	Raised over period				\$2.80B	Raised over period				\$2.00B
Dilution by FY36				+74%	Dilution by FY36				+27%	Dilution by FY36				+10%	Dilution by FY36				+18%
FY36 shares (M)				3,401	FY36 shares (M)				2,480	FY36 shares (M)				2,145	FY36 shares (M)				2,010
DCF per share				\$0.19	DCF per share				\$3.87	DCF per share				\$42.28	DCF per share				\$21.37
× (1-P_fail) [55%]				\$0.10	× (1-P_fail) [75%]				\$2.90	× (1-P_fail) [88%]				\$37.21	× (1-P_fail) [97%]				\$20.73
+ P_fail × distress [45% × \$0.00]				\$0.00	+ P_fail × distress [25% × \$0.50]				\$0.13	+ P_fail × distress [12% × \$1.00]				\$0.12	+ P_fail × distress [3% × \$0.98]				\$0.03
= Expected per share				\$0.11	= Expected per share				\$3.03	= Expected per share				\$37.32	= Expected per share				\$20.76
FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT				
+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y	
\$0.18	\$0.29	\$0.46	\$0.74		\$4.56	\$6.85	\$10.30	\$15.49		\$53.58	\$76.92	\$110.43	\$158.53		\$30.50	\$44.82	\$65.85	\$96.76	
0.02×	0.04×	0.06×	0.10×		0.59×	0.89×	1.33×	2.00×		6.92×	9.94×	14.27×	20.48×		3.94×	5.79×	8.51×	12.50×	



Aurora Innovation supporting analysis · disclaimers · glossary

Bear \$0.11 Base \$3.03 Bull \$37.32

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

- 1 **Commercial operation is real.**
Aurora Driver runs supervised highway pilots with McLane (Berkshire), PACCAR, Volvo. Removes 'science project' bear framing.

4 **First-mover regulatory moat.**
Aurora has logged the most miles in commercial operation. Lane data, TX/OK/NM regulatory relationships compound.
- 2 **Partnership stack across OEMs.**
Volvo VNL Autonomous and PACCAR Peterbilt/Kenworth integrations. Unlike Kodiak (cargo-only) or Waabi (Volvo-only), Aurora spans both.

5 **Mature margin may be conservative.**
Base 20% reflects tech-auto blend. If software-take-rate on OEM trucks scales (Mobileye analog), terminal could be 30%+.
- 3 **\$1.28B liquidity + capital flexibility.**
Q1 2026 raise extended runway. Strategic options (M&A interest from majors) provide downside floor.

FALSIFICATION TRIGGERS

Bear validation Driverless launch slips to 2028+ · Cash < \$500M by mid-2027 · Commercial volume disappoints (< 100 trucks by EOY 2028)	Bull validation Driverless commercial launches by Q3 2027 · Fleet > 500 trucks by EOY 2028 · Second OEM software-take-rate deal signed	Reframe needed If Kodiak achieves at-scale commercial first — Aurora's first-mover advantage in long-haul erodes; TAM-share math compresses
---	--	---

DISCLAIMERS · PLEASE READ BEFORE USING THIS DOCUMENT

Not investment advice. This document is produced for entertainment and educational purposes only by an individual amateur investor. "Alameda Research 2: Electric Boogaloo" is the unincorporated personal concept of one person and is not a registered investment management entity. Nothing herein constitutes investment advice, a recommendation, or a solicitation to buy or sell any security.	Not a professional. The author is not a registered investment advisor, broker-dealer, CFA charterholder, certified financial planner, or licensed financial professional of any kind. The author has no formal training in equity research or capital markets. No fiduciary relationship is created by reading this document.	AI-assisted analysis. Substantial portions of this analysis were produced with the assistance of large language models. LLMs are known to fabricate facts, miscalculate numbers, hallucinate sources, and present incorrect information with high apparent confidence. Every figure, claim, and projection in this document should be independently verified before acting on it.
Forward-looking statements. Scenarios, valuations, projections, and any forward-looking statements involve substantial assumptions and uncertainty. Actual results may differ materially from those projected. Past performance is not indicative of future results. The model is a model; reality is not.	Author may hold positions. The author may own, intend to acquire, or hold short exposure to \$AUR or related securities at any time. Positions may change without notice and without an update to this document. Do not assume the author's positions match the tone of the analysis.	Do your own research. Do not make investment decisions on the basis of this document. Consult a licensed financial professional and read the company's official SEC filings (10-K, 10-Q, 8-K, proxy) before forming any investment view. If you wouldn't trust your retirement to a chatbot, don't trust this either.

GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Aurora Driver — Aurora's autonomy software + hardware stack. Currently runs supervised on highway pilots with PACCAR, Volvo, and FedEx.	Driverless commercial — Operation with no human driver in the truck, generating per-mile or per-load revenue. Aurora's 2027 launch milestone.	S2C (sales-to-capital) — Damodaran reinvestment ratio. $\Delta \text{Revenue} / \text{S2C}$ = reinvestment in \$. Higher = more capital-efficient growth.
Hub-to-hub — Highway-only routes between terminals (e.g., Dallas-Houston). Aurora's commercial focus. Easier autonomy domain than urban streets.	OEM partners — Volvo VNL Autonomous and PACCAR (Peterbilt, Kenworth). OEM integration gives Aurora a manufacturing path competitors lack.	Take-rate model — Revenue per autonomy-mile on OEM-built trucks (vs Aurora-owned fleet). Software-like margins; the bull terminal-margin driver.